Jun Du

(346) 978-6515 — justindujun@gmail.com — GitHub — Personal Website

Education

Texas A&M University College Station, TX
B.S. in Computer Science *Expected May 2027*
- GPA: 3.71 — Texas Leadership Scholar — Regent Scholar — Houston Livestock Show & Rodeo Scholarship

Experience

Platform Engineer Co-op Incoming – Fall 2026
RTX (Raytheon) *Richardson, TX*
- Selected for a platform engineering co-op supporting internal developer platforms and infrastructure tooling for defense systems software teams. Getting Top Secret for clearance.

Data Engineer Intern May 2026 – Aug 2026
AST SpaceMobile *Miami, FL*
- Built an internal battery test-data dashboard (Next.js 16, React 19, TypeScript, Tailwind 4) with interactive Recharts visualizations and an Excel-ingestion view linking satellite-module assembly records to battery packs, enabling real-time monitoring and failure analysis for engineering teams.
- Engineered server-rendered ETL data pipelines with Prisma 7 over a remote MariaDB, writing optimized SQL aggregate queries (subquery-based filtering, request-level caching) to power browsable inventory tables and cell/pack lifecycle views.
- Developed a network-resilience layer (per-attempt timeouts, transient-error retries, compression, automatic VPN-to-RDS failover) that eliminated intermittent query hangs and connection resets over an unstable database path.
- Designed and built a custom relay PCB for the battery Functional Test (FT) station, enabling 4 distinct test procedures to run on a single fixture and tripling technician throughput from 50 to 150 batteries per 8-hour shift.

Autonomous Software Engineer March 2026 – Present
Texas A&M Vertical Flight Design *College Station, TX*
- Develop autonomy software for the team's uncrewed vertical-flight aircraft competing in the 2026 Student Unmanned Aerial Systems (SUAS) competition.
- Authored a competition requirements document derived from the SUAS Team Handbook and delivered a tiltrotor configuration trade study to inform vehicle design decisions.

Undergraduate Research Assistant – Computer Vision Aug 2025 – Dec 2025
Texas A&M Hazard Reduction & Recovery Center *College Station, TX*
- Trained YOLOv11 object detection models (PyTorch) to identify physical disorder indicators (litter, graffiti, abandoned buildings) in post-hurricane street-level imagery, using 3,000+ images with 5,000+ annotations.
- Built the end-to-end ML pipeline from data labeling and annotation through model training on NVIDIA A100 GPUs, collaborating with urban planning and public health researchers to translate outputs into disaster recovery insights.

Machine Learning & AI Intern May 2025 – Aug 2025
Crux AI *College Station, TX*
- Architected and deployed an end-to-end computer vision system (PyTorch, YOLOv11) combining object detection with pose estimation to reconstruct climbing movement from video into structured metrics (hold transitions, joint kinematics, trajectories).
- Curated and annotated 7,000+ domain-specific images of climbing holds; trained and optimized custom detection models achieving ∼95% accuracy under varied lighting and route conditions.
- Deployed Dockerized inference services on Microsoft Azure Container Apps, applying MLOps practices to serve ML models efficiently at scale.
- Integrated a generative-AI (LLM) analysis layer, applying prompt engineering to generate strategy-level feedback on climbing efficiency, route optimization, and technique.

Projects

Maroon Ride | *React Native, TypeScript, Mobile Development* Jan 2024 – May 2024
- Contributed to a campus ride-sharing app with 10,000+ downloads on the Apple App Store; implemented ETA logic and backend route-planning improvements for real-time ride coordination.

Leadership & Activities

Organizations | *Aggie Coding Club, Data Science Club, Data Science Hackathon* Texas A&M University

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, Java, C#, C++, HTML/CSS
AI / ML: PyTorch, Computer Vision, YOLOv11, OpenCV, Machine Learning, LLM Integration, Prompt Engineering, Generative AI
Web & Frameworks: React, Next.js, Node.js, React Native, Flutter, Prisma
Data & Cloud: ETL/Data Pipelines, Real-time Analytics, MariaDB, MySQL, AWS, Azure, Google Cloud Platform, Docker, Kubernetes, CI/CD, Linux (Ubuntu)
Tools & Libraries: Git, Azure DevOps, Pandas, NumPy, Matplotlib, Recharts, Roboflow
Languages Spoken: English, Mandarin Chinese